



RAPHAEL PENZ

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SUMMARY

Games Engineering graduate from the Technical University of Munich with experience developing and releasing a large-scale solo RPG/Visual Novel project. Strong background in gameplay systems, software architecture, Python, and C++, complemented by experience in high-performance simulation development through academic research. Interested in gameplay programming, systems design, and software development roles.

PROJECT EXPERIENCE

Harmony of the Stars

(Game Project)

Sept 2024 - Present

- **Solo Developer** | theh0bbit.github.io/
 - Designed and developed a solo RPG / Visual Novel hybrid using Python and Ren'Py.
 - Implemented custom gameplay frameworks including combat, inventory, quests, progression, relationships, scheduling, and world navigation.
 - Developed approximately 23,000+ lines of gameplay code and 47,000+ lines of total project code.
 - Released public builds and maintained compatibility across updates and save files.
 - Achieved more than 2,000 downloads across public releases.
 - Solely responsible for programming, systems design, project architecture, worldbuilding, and release management.

Implementation and

Evaluation of

Additional Particle

Simulation Types with

AutoPas

(Bachelor Thesis)

October 2021

- **Technical University of Munich**
 - Implemented a new Discrete Element Method (DEM) particle type in the AutoPas C++ particle simulation framework.
 - Developed pairwise force calculation functors supporting AoS and SoA memory layouts with SIMD optimization.
 - Extended simulation infrastructure and integrated DEM simulations into the existing molecular dynamics benchmark suite.
 - Evaluated scalability and performance on simulations containing up to 1,000,000 particles.
 - Compared physical behavior and runtime characteristics against existing molecular dynamics implementations.

EDUCATION

**M.Sc. Computer
Science: Games
Engineering**

- **Technical University of Munich (TUM)**
Munich • Oct 2024 - Sept 2025
• Completed 90/120 ECTS

**B.Sc. Computer
Science: Games
Engineering**

- **Technical University of Munich (TUM)**
Munich • Oct 2017 – Sept 2024

WORK EXPERIENCE

**Hotel Assistant (Part-
Time)**

August 2021 - Present

- **Luxury Boutique Hotel Himmlhof** | St. Anton am Arlberg, Tirol
 - Supported day-to-day hotel operations in a part-time capacity.
 - Assisted with guest-facing services, reception support, housekeeping, and general operational tasks.
 - Adapted to varying responsibilities depending on seasonal and operational requirements.

**Civilian Service /
Emergency Medical
Technician (EMT)**

September 2016 - June 2017

- **Austrian Red Cross** | Landeck, Tirol
 - Served as a paramedic
 - Supported dispatch operations, including call handling, scheduling patient transport services, and coordinating drivers and vehicles.

SKILLS

Programming Languages

C++, Python, C#, Java

Software Development

Git, Object-Oriented Design, Gameplay Systems Development, Data-Driven Development, Save & Update-Compatible Architecture

Tools & Engines

Git, Ren'Py (Advanced), Unity (Intermediate), Unreal Engine (Basic Familiarity), Gimp, Photoshop, DaVinci Resolve, Audacity

Languages

German (Native), English (C2 Proficient)